Preprint

LETTER

The Other Half of Gravity: Torsion and the Two-Chamber Structure of the Tetrad Frame

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Abstract
The connection on the tetrad frame has $K^3 = 64$ independent components. General relativity's zero-torsion condition removes 24 of them, the antisymmetric sector, retaining 40 Christoffel symbols determined by the metric. The companion letter [3] proved that the tetrad equation produces two bounded tetrahedral regions related by spatial parity: TetA (the identity component) and TetB (the parity component). The present letter proves that a torsion-free connection cannot distinguish TetA from TetB (Lemma, Section 2): the torsion sector is the necessary carrier of the two-chamber distinction, and the zero-torsion condition collapses it. The exchange-symmetric partition derived in Supplementary Note 1 uniquely determines the TetB field fraction as $1/(x{+}1)$, where $x = g/a_0$ is the dimensionless acceleration. A coupling polynomial $f(K, s) = 1 + sK + K^2$ organises the full count structure and yields the torsion identity $24 = D \times 2K$ exclusively at $K = 4$ (Section 5). The ratio of the removed to the retained field response is $1/x$, which is algebraically identical to the dark-to-baryon mass ratio of the cold dark matter framework.

1 The torsion question

The vielbein formulation of general relativity describes the gravitational field through a frame field $e^a{}_\mu$ with $K = 4$ legs [1]. The affine connection $\Gamma^I{}_{JK}$ on this frame carries three frame indices and therefore has $K^3 = 64$ independent components. Under exchange of the lower index pair, these components divide into two families: 40 that are symmetric under the exchange, and 24 that are antisymmetric. The symmetric components are the Christoffel symbols, fully determined by the metric and its first derivatives. The antisymmetric components are the torsion [2].

In 1915, Einstein formulated general relativity with the assumption that the connection is symmetric in its lower indices, setting all 24 torsion components to zero. This was recognised early as a simplifying choice rather than a geometric necessity. Cartan [4] showed in 1922 that the mathematical framework accommodates torsion naturally, and Hehl and collaborators [2] developed the full Einstein-Cartan theory, in which the 24 torsion components couple to the spin density of matter. The resulting theory reproduces every prediction of general relativity in vacuum and differs only inside extremely dense spinning matter.

Yet the geometric identity of the 24 components, what they represent in the structure of the tetrad frame, beyond their index antisymmetry, has remained an open question. The companion letter [3] provides the tools to answer it. That letter proved that the tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ produces two bounded tetrahedral regions in its spatial sector, related by spatial parity $P : e^i \to -e^i$. The identity component gives TetA; the parity component gives TetB. Their union is the stellated octahedron.

The present letter identifies the 24 torsion components with TetB, derives the fraction of the gravitational field response they carry, and traces the consequences of their removal.

2 The parity identification

The argument requires two established results and one lemma that connects them.

The first result is geometric. The companion letter [3] proved that spatial parity decomposes the tetrad equation's spatial sector into two bounded regions: TetA, which is the identity component under $P : e^i \to -e^i$, and TetB, which is the parity component. The identification of vielbein legs with oriented tetrahedral faces is established in that companion letter and is used here only as prior geometric input.

The second result is algebraic and entirely standard. The affine connection $\Gamma^I{}_{JK}$ decomposes under exchange of the lower index pair $(J \leftrightarrow K)$ into two sectors: the 40 symmetric components, which are invariant under the exchange, and the 24 antisymmetric components, which reverse sign [1, 2]. The symmetric sector is the identity sector under index exchange; the antisymmetric sector is the parity sector.

The following lemma connects the two decompositions.

Lemma. *A torsion-free connection cannot distinguish TetA from TetB.*

Proof. A torsion-free connection is uniquely determined by the metric and its first derivatives

(the fundamental theorem of Riemannian geometry [1]). The metric $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ is bilinear in the frame field. Under the spatial parity $P : e^i \to -e^i$ that exchanges TetA and TetB, both factors acquire a sign flip; the two signs cancel, leaving $g_{\mu\nu}$ invariant. The metric is therefore invariant under TetA $\leftrightarrow$ TetB. A torsion-free connection, being fully determined by this invariant metric, inherits that invariance: it treats TetA and TetB identically. □

Corollary. The torsion sector is the necessary carrier of the TetA/TetB distinction at the connection level. Without torsion, the connection is built entirely from the parity-invariant metric and cannot encode which of the two tetrahedral chambers a given field response belongs to. Setting torsion to zero does not merely reduce the number of connection components from 64 to 40; it collapses the two-chamber structure of the tetrad geometry into a single effective sector.

The torsion-free specialisation $\Gamma^I{}_{[JK]} = 0$ therefore sets to zero the only sector of the connection that can carry the TetA/TetB distinction. In the partition language developed below, this means attributing the full field response to a single effective sector.

3 The TetB field fraction

Because TetA and TetB together exhaust the bounded structure of the tetrad equation, the gravitational field response at each point in space partitions between them. Let $\mu(x)$ denote the fraction of the total field response carried by TetA at dimensionless acceleration $x = g/a_0$, so that $1 - \mu(x)$ is the fraction carried by TetB. The two fractions sum to unity because μ measures the relative partition between the two coupled sectors, not the absolute field strength. In the weak-field regime relevant to galactic dynamics, the connection components that govern test-particle motion reduce to gradients of the Newtonian potential, so the field-response partition acts directly on the gravitational acceleration.

Spatial parity exchanges TetA and TetB. Under this exchange the acceleration ratio inverts and the two fractions swap, imposing the functional equation $\mu(1/x) = 1 - \mu(x)$. Supplementary Note 1 proves that this exchange symmetry, combined with the requirement that the partition be a rational bijection on the projective line (a classification theorem of Hartshorne [5]), uniquely determines the TetA and TetB fractions:

$$\text{TetA:} \quad \mu(x) = \frac{x}{x+1}, \tag{1}$$

$$\text{TetB:} \quad 1 - \mu(x) = \frac{1}{x+1}. \tag{2}$$

The derivation uses no field equation, no coupling polynomial, and no fitting. It requires only the parity relation between TetA and TetB proved in the companion letter together with the rational-bijection classification stated above.

Setting torsion to zero removes TetB from the connection. In the partition language, this sets $1 - \mu = 0$ and therefore $\mu = 1$: the full gravitational field response is attributed to TetA alone. The exchange symmetry proves that $\mu \neq 1$. The field response absent from the torsion-free branch is

$$\Delta\mu(x) = \frac{1}{x+1}. \tag{3}$$

The consequences of this removal depend on the acceleration regime. At high acceleration ($x \gg 1$), the TetB fraction is negligible: $\Delta\mu \approx 0$, and general relativity is accurate. At the parity midpoint ($x = 1$), TetA and TetB contribute equally: $\mu = 1/2$ and $\Delta\mu = 1/2$, so that the torsion-free branch does not capture half of the gravitational field response. At low acceleration ($x \ll 1$), the TetB fraction approaches unity: $\Delta\mu \approx 1$, and the torsion-free branch does not capture nearly any of it.

Regime	x	μ (retained)	$\Delta\mu$ (removed)
Strong field	$\gg 1$	≈ 1	≈ 0
Transition	$= 1$	$1/2$	$1/2$
Weak field	$\ll 1$	≈ 0	≈ 1

4 The dark-to-baryon ratio

The cold dark matter framework accounts for the discrepancy between observed gravitational fields and baryonic predictions by introducing an invisible mass component. Its decomposition of the observed field is [6]:

$$g_{\text{obs}} = g_{\text{baryon}} + g_{\text{DM}}, \tag{4}$$

where g_{DM} is the residual: whatever the observed field contains beyond the baryonic contribution. This is the definition of the dark matter component.

In the partition language developed above, the baryonic contribution is the TetA share of the observed field, $g_{\text{baryon}} = \mu\, g_{\text{obs}}$, and the residual is the TetB share, $g_{\text{DM}} = (1 - \mu)\, g_{\text{obs}}$. The ratio of the residual to the baryonic contribution is therefore:

$$\frac{g_{\text{DM}}}{g_{\text{baryon}}} = \frac{1-\mu}{\mu} = \frac{1/(x+1)}{x/(x+1)} = \frac{1}{x}. \tag{5}$$

This is an algebraic identity of the partition, not an empirical fit. The cold dark matter framework defines its invisible component as the residual after subtracting the Newtonian contribution. The exchange-symmetric partition derived in Section 3 determines that residual as $1/(x+1)$. Because the effective mass at radius r

is $M = gr^2/G$ for any acceleration g, the acceleration ratio at each shell is identically the enclosed-mass ratio. The resulting dark-to-baryon mass ratio,

$$\boxed{\frac{M_{\rm DM}}{M_{\rm baryon}} = \frac{1}{x},} \tag{6}$$

follows from four lines of algebra with no halo profile and no free parameters. To be explicit: the ratio $1/x$ is not an independent quantity. It is the direct consequence of the constitutive partition $\mu(x) = x/(x+1)$ derived in Section 3. Wherever TetA carries a fraction μ of the field, TetB carries $1-\mu$, and their ratio is $(1-\mu)/\mu = 1/x$. The cold dark matter framework requires a multi-parameter halo profile, fitted separately for each galaxy, to obtain this ratio. The exchange-symmetric partition of the two tetrahedral chambers reduces it to the single function $1/x$, with no free parameters. The torsion-sector field fraction and the cold-dark-matter residual are algebraically identical under the exchange-symmetric partition: both reduce to $1/(x{+}1)$, with the corresponding dark-to-baryon ratio $1/x$.

5 The coupling polynomial

The parity identification in Sections 2–4 relies on the decomposition of the connection into 40 symmetric and 24 antisymmetric components, which is textbook. A natural question is why these particular numbers arise. The answer is that both, together with every other standard count of the tetrad formalism, are polynomial functions of a single integer $K = 4$.

The tetrad equation contains three structural ingredients at ranks 0, 1, and 2 on the frame space, with graded dimensions $K^0 = 1$, $K^1 = K$, and K^2. The companion letter [3] established the ordered three-region structure TetA, the interaction vertex, and TetB. The present letter encodes those three regions by the state label $s \in \{+1, 0, -1\}$.

The identification of vielbein legs with oriented tetrahedral faces is proved in that companion letter and is used here only as prior input for the structural labelling. Since the rank-1 contribution is parity-odd while the rank-0 and rank-2 contributions are parity-even, the state label s modulates only the linear term, yielding the coupling polynomial:

$$f(K, s) = 1 + sK + K^2. \tag{7}$$

At $K = 4$, the three evaluations are $f = 21$, 17, and 13 for $s = +1$, 0, and -1 respectively, with a coupling range of $\Delta f = 2K = 8$. At the interaction vertex ($s = 0$), which is the non-oriented coupling locus where TetA and TetB meet, the parity-odd directional contribution is absent.

The polynomial organises the full count structure of the tetrad formalism. In the broader framework developed in the companion letter, the three evaluations {21, 17, 13} are not merely bookkeeping values but coupling-depth reads of the three structural states; the present letter establishes the polynomial and its concordances, with the fuller physical interpretation developed there.

The standard symmetric and antisymmetric counts of the connection follow from K alone:

$$\text{Christoffel:} \quad K \times \frac{K(K{+}1)}{2} = 40, \tag{8}$$

$$\text{Torsion:} \quad K \times \frac{K(K{-}1)}{2} = 24. \tag{9}$$

The torsion count admits an additional polynomial factorisation that connects it to the coupling range.

Theorem. *Under the closure relation $D = K - 1$, the torsion count $K \times K(K{-}1)/2$ equals the product $D \times 2K$ if and only if $K = 4$.*

Proof. Set the two expressions equal and divide by $K > 0$: $K(K{-}1)/2 = 2D$. Substitute $D = K - 1$: $K(K{-}1)/2 = 2(K{-}1)$. For $K \neq 1$, divide by $(K{-}1)$:

$$\frac{K}{2} = 2 \qquad \Longrightarrow \qquad K = 4. \qquad \square \tag{10}$$

The 24 torsion components that carry the TetB field fraction therefore admit the factorisation $D \times \Delta f = 3 \times 8$: the product of the spatial dimension and the coupling range. This factorisation holds at no other positive integer.

Once the standard tetrad counts are specified, they admit a unified polynomial organisation in K:

Standard tetrad counts		
Scalar closure	1	K^0
Vielbein legs	4	K
Metric entries	16	K^2
Symmetric metric	10	$K(K{+}1)/2$
Antisym. pairs	6	$K(K{-}1)/2$
Connection	64	K^3
Christoffel	40	$K \times K(K{+}1)/2$
Torsion	24	$K \times K(K{-}1)/2$
Polynomial evaluations (this letter)		
TetA depth ($s{=}{+}1$)	21	$1 + K + K^2$
Core depth ($s{=}0$)	17	$1 + K^2$
TetB depth ($s{=}{-}1$)	13	$1 - K + K^2$
Coupling range	8	$2K$

The polynomial separates state-invariant structure from state-sensitive structure: the scalar closure term K^0 and the bilinear coupling term K^2 persist across all three structural

states, while only the rank-1 contribution changes with s. The polynomial therefore provides a compressed classifier and projection rule for the frame construction: $s = 0$ isolates the parity-even core, while $f(K, +1) - f(K, -1) = 2K$ extracts the orientation-carrying sector.

6 Discussion

This letter rests on two independent arguments. The first is the parity identification (Sections 2–4): the torsion sector is the necessary carrier of the TetA/TetB distinction, the exchange-symmetric partition derived in the supplementary note gives the TetB fraction as $1/(x{+}1)$, and the dark-to-baryon ratio follows algebraically. This argument requires only the companion geometry and textbook general relativity. The second is the polynomial organisation (Section 5): the coupling polynomial generates the standard counts, and the torsion identity links the antisymmetric count to the coupling range at $K = 4$. A challenge to one argument does not affect the other. Just as the metric and Christoffel symbols are kinematic objects determined by the tetrad equation without reference to an action principle, the TetA/TetB partition is determined by its parity structure; the result therefore holds independently of the specific gravitational dynamics chosen.

Einstein-Cartan theory retains the 24 torsion components and couples them to the spin density of matter [2]. In the present framework, retaining the torsion means retaining TetB in the connection and preserving the full field partition $\mu(x) = x/(x{+}1)$. Cartan's instinct that torsion belongs in the geometry finds a new motivation here: the torsion sector carries the field response of the second tetrahedral chamber, without which the connection describes only half of the gravitational field at the parity midpoint.

At high acceleration ($x \gg 1$), the torsion correction $\Delta\mu = 1/(x{+}1)$ is negligible. General relativity's predictions in the strong-field regime (solar system dynamics, binary pulsar timing, gravitational wave propagation) are unaffected. The torsion fraction becomes significant only at accelerations of order a_0 and below, which is precisely where the cold dark matter framework introduces its invisible component. The same saturation resolves a longstanding internal tension in that framework: because the TetB field fraction $1/(x{+}1)$ is bounded above by unity, the corresponding field acceleration never exceeds a_0, forcing a finite central density rather than the divergent cusp that an NFW profile requires [6].

The function $\mu(x) = x/(x + 1)$ was identified empirically by Milgrom [7] as an interpolating function for galactic dynamics and selected by fitting by Famaey and Binney [8]. The present derivation obtains it from the exchange symmetry of the two parity-related bounded regions of the tetrad equation, without fitting and without reference to galactic data.

The bimetric gravity programme [9] independently features $2K = 8$ as the total vielbein count of two interacting tetrad fields and a $K \times K$ interaction structure, giving $K^2 = 16$ at $K = 4$. The closure condition $2(D{+}1) = 2^D$ with $D = K - 1$ exhibits a striking concordance across three layers of the tetrad formalism: at the geometric level, it selects the stellated octahedron as the unique closure geometry [3]; at the algebraic level, it produces the numerical coincidence $K^3 = K^D$; and at the connection-theoretic level, it is the unique value at which the torsion count admits the polynomial factorisation $D \times 2K$.

The torsion sector carries the gravitational field response of the second tetrahedral chamber. The torsion-free specialisation adopted in 1915 does not carry it.

7 Data and reproducibility

The exchange-symmetry derivation of the TetB field fraction is provided as a supplementary note (`supplementary_tetb_fraction.pdf`). A self-contained validation script (`run_all_validations.py`) verifies every claim in both this letter and its companion [3], requiring only NumPy, SciPy, and SymPy. Sections 1–12 of the script verify the geometric claims of the companion letter; Section 13 verifies the Lemma of the present letter by confirming $P^T \eta\, P = \eta$ for all component sectors (g_{00}, g_{0i}, g_{ij}); Section 14 verifies the algebraic identity chain from the exchange symmetry through the dark-to-baryon ratio, using symbolic computation with zero numerical tolerances (52/52 claims passed).

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